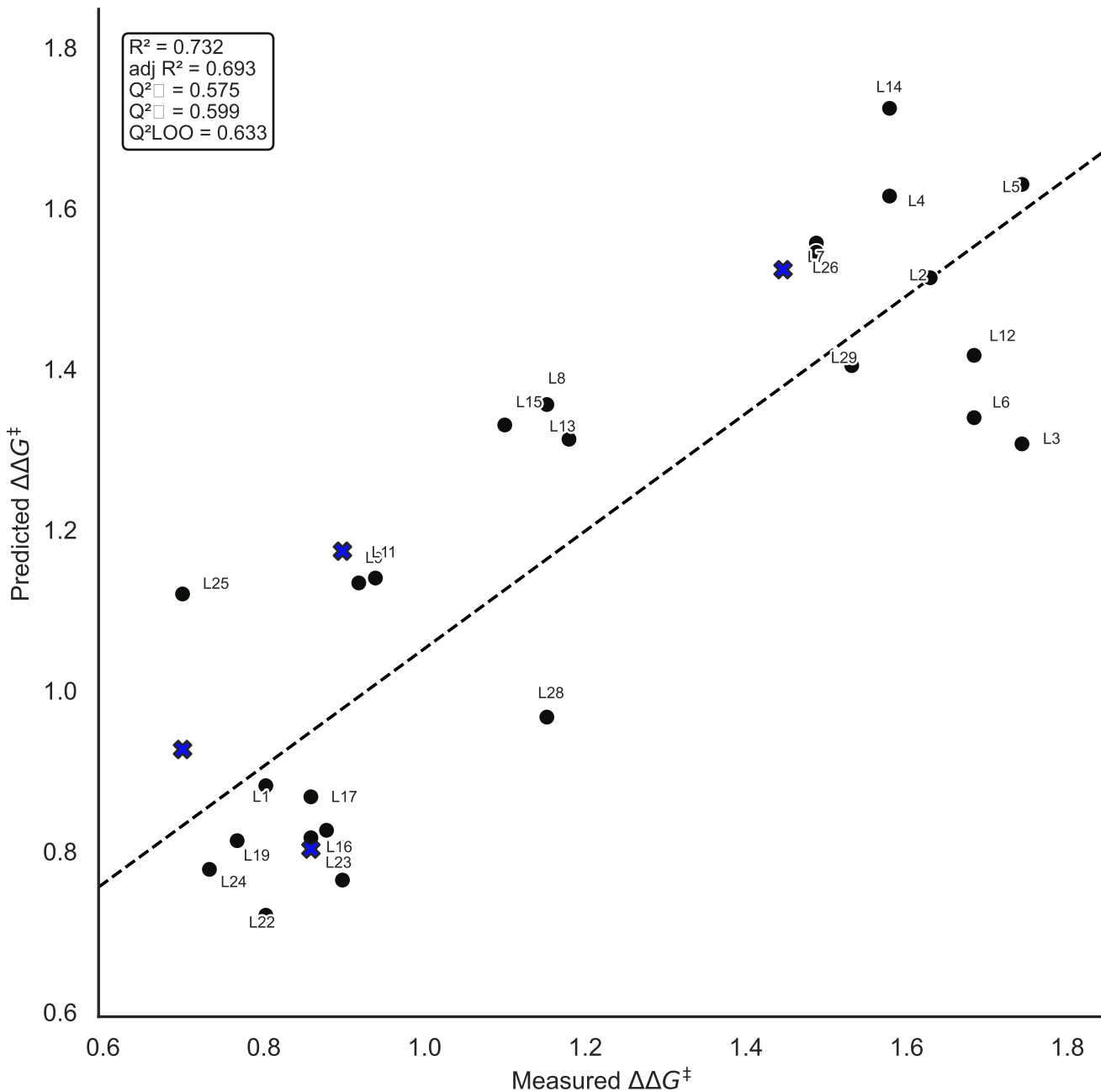


Predicted vs Measured



Model Metrics

Metric	Value
R^2 (train)	0.7316
adj R^2 (train)	0.6933
MAE (train)	0.1518
Q^2 3-fold	0.5755
MAE 3-fold	0.1862
Q^2 5-fold	0.5991
MAE 5-fold	0.1877
Q^2 LOOCV	0.6325
MAE LOOCV	0.1796
R^2 (ext. val.)	0.5684
adj R^2 (ext. val.)	nan
MAE (ext. val.)	0.1593

Coefficient Estimates

Estimate	Std. Error	t value	p value
1.183	0.042	28.472	0.0
0.157	0.04	3.916	0.001
0.228	0.044	5.194	0.0
0.119	0.044	2.726	0.013

Cross-Validation Metrics

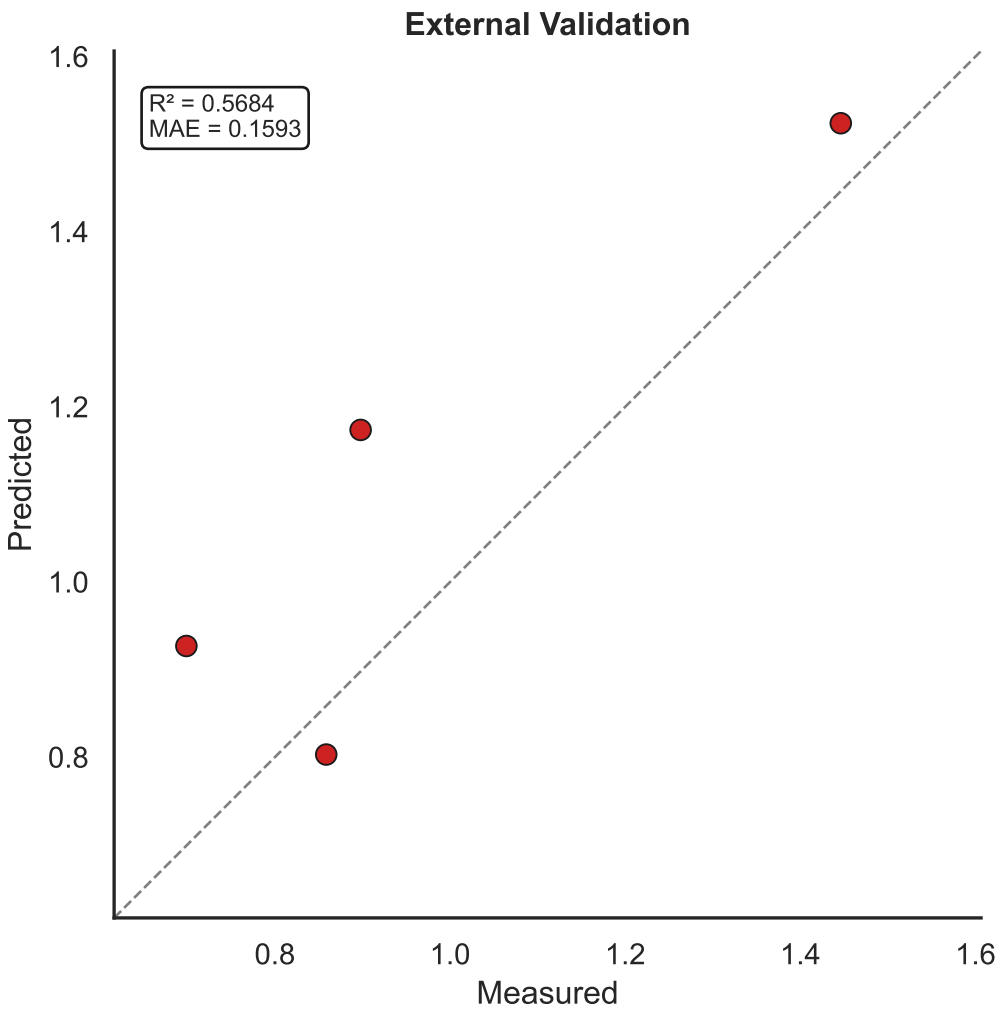
Metric	3-Fold	5-Fold	LOOCV
Q²	0.5755	0.5991	0.6325
MAE	0.1862	0.1877	0.1796
RMSD	0.2386	0.2318	0.2225

Variance Inflation Factors (VIF)

variable	VIF
energy	1.087
B1_B5_angle_5-2	1.066
dihedral_[5, 1, 8, 6]	1.063

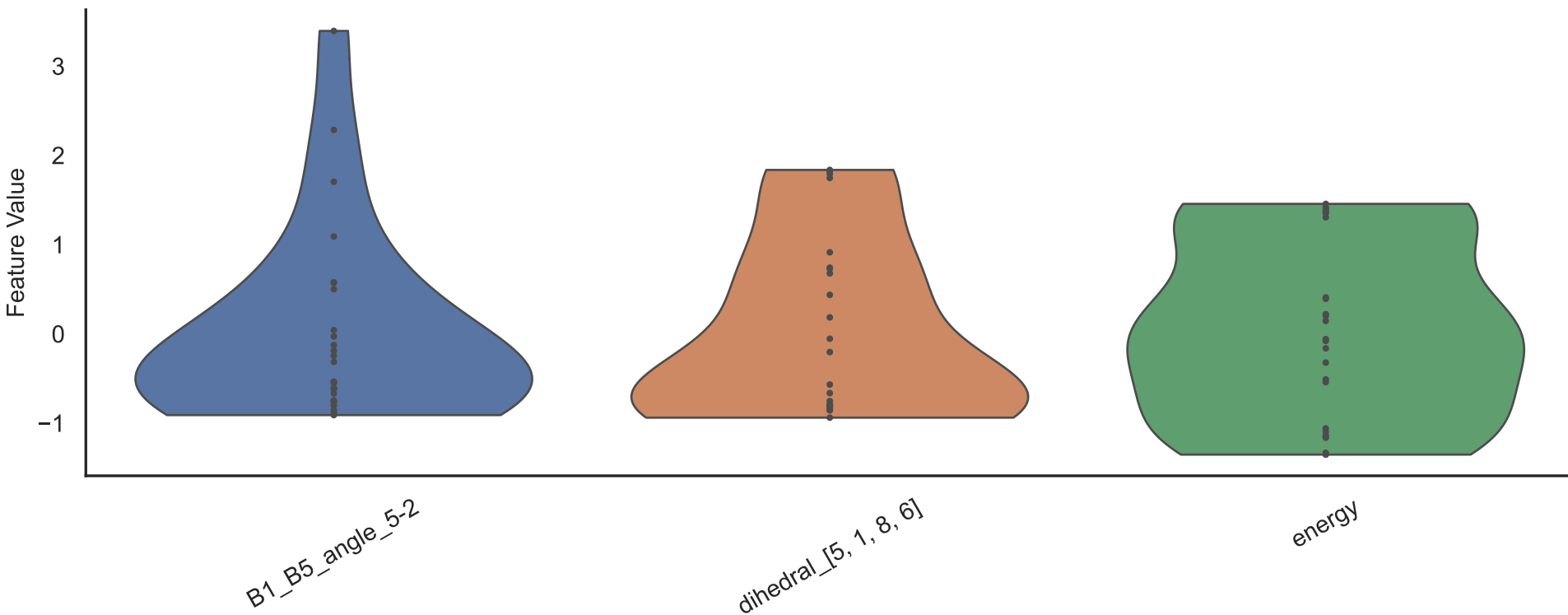
Coefficient Estimates

Estimate	Std. Error	t value	p value
1.1832	0.0416	28.4716	0.0
0.1575	0.0402	3.916	0.0008
0.2277	0.0438	5.1943	0.0
0.1194	0.0438	2.7263	0.0127



Molecule	Actual	Predicted	Error in %
L10	0.898	1.174	30.711
L27	1.446	1.524	5.369
L21	0.699	0.927	32.626
L18	0.859	0.803	6.482

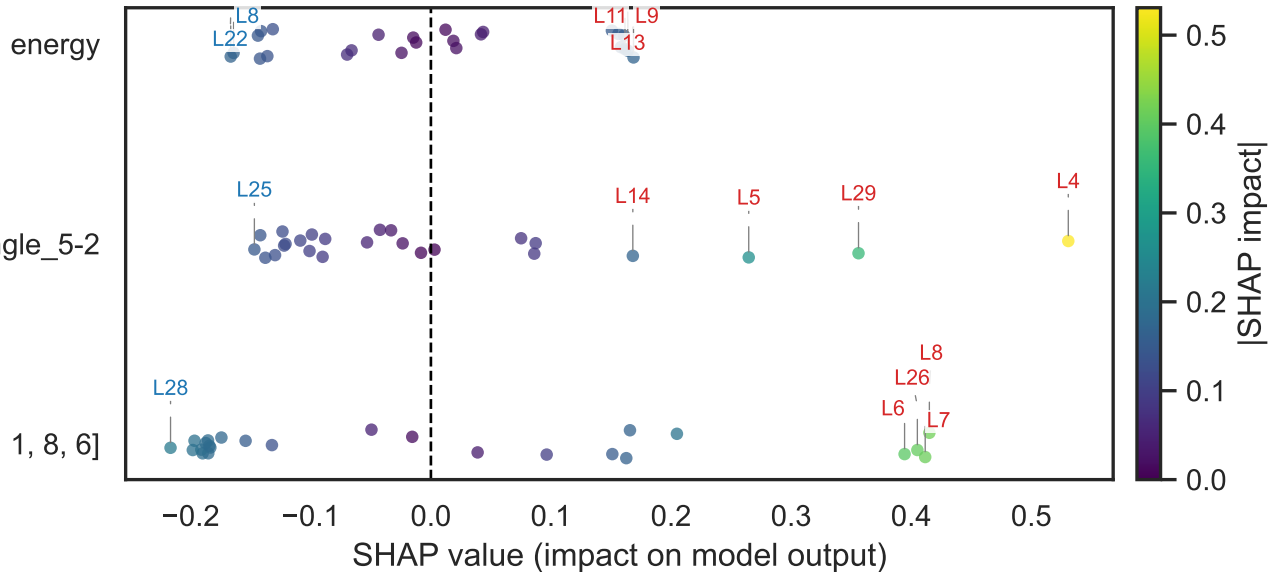
Distribution of Selected Features



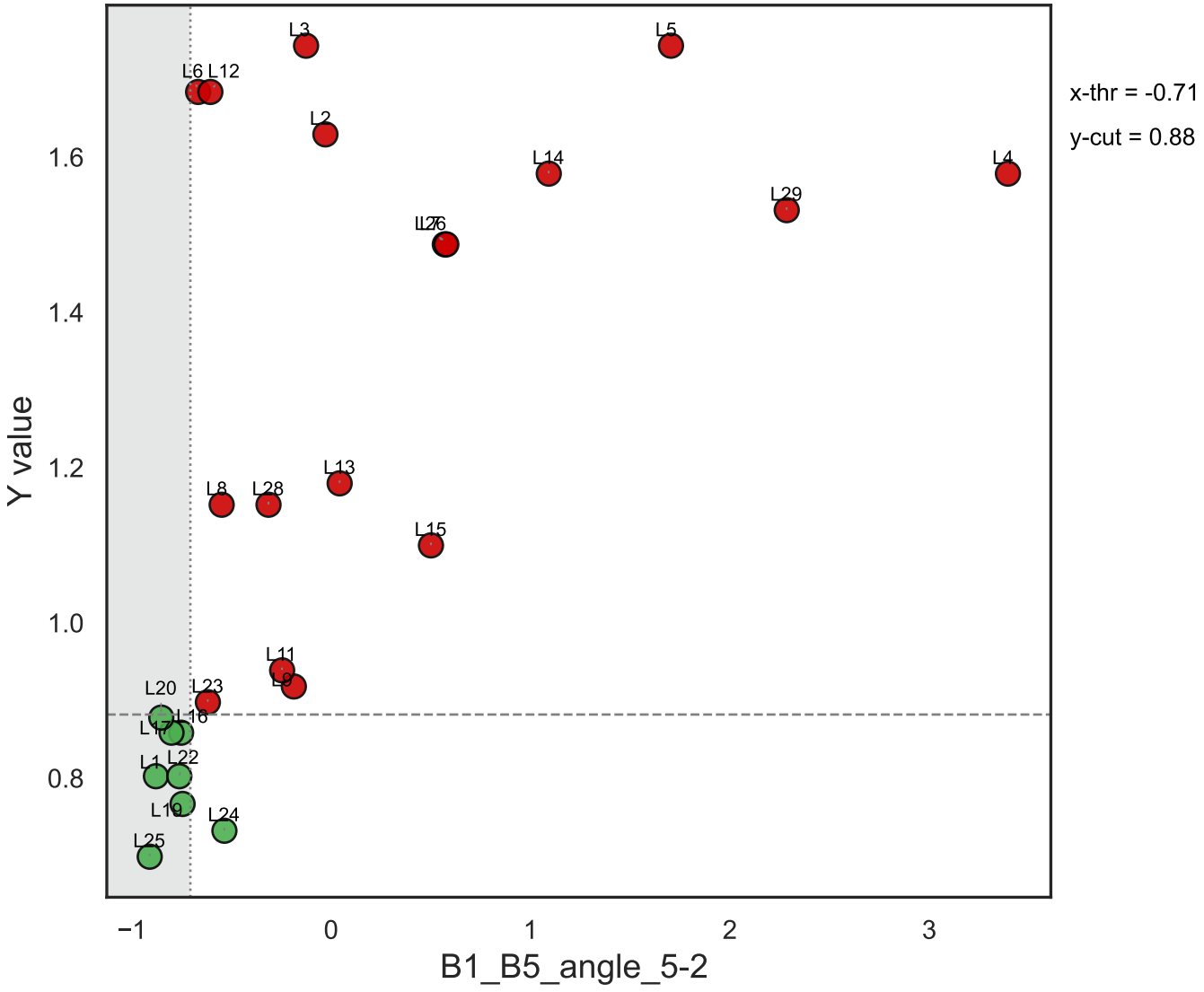
Descriptive Statistics

Feature	mean	std	min	25%	50%	75%	max
B1_B5_angle_5-2	0.024	1.088	-0.91	-0.745	-0.314	0.501	3.394
dihedral_[5, 1, 8, 6]	0.014	0.996	-0.938	-0.801	-0.567	0.729	1.837
energy	0.044	1.008	-1.352	-1.059	-0.059	1.306	1.457

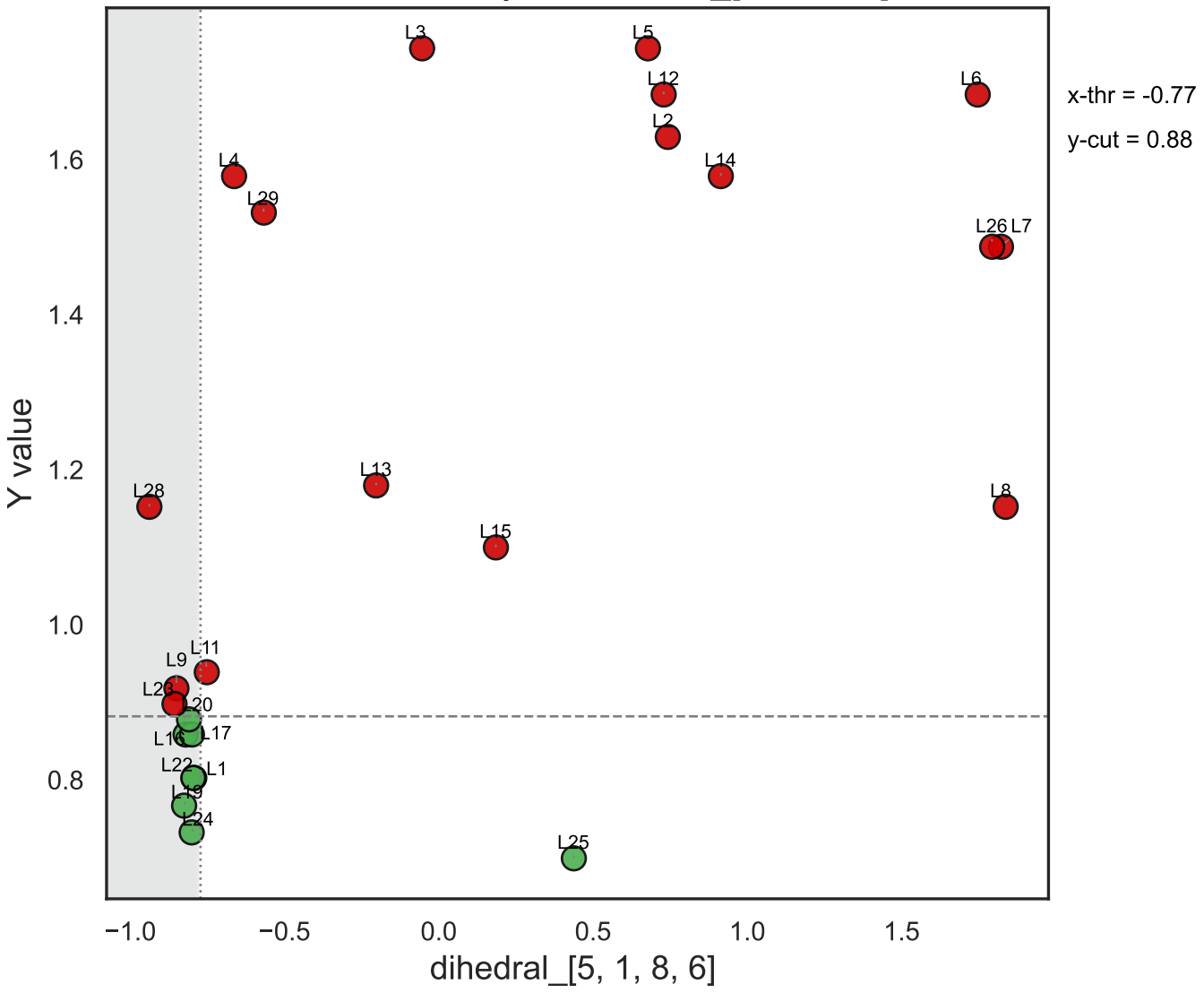
SHAP beeswarm — top 3 features (target)



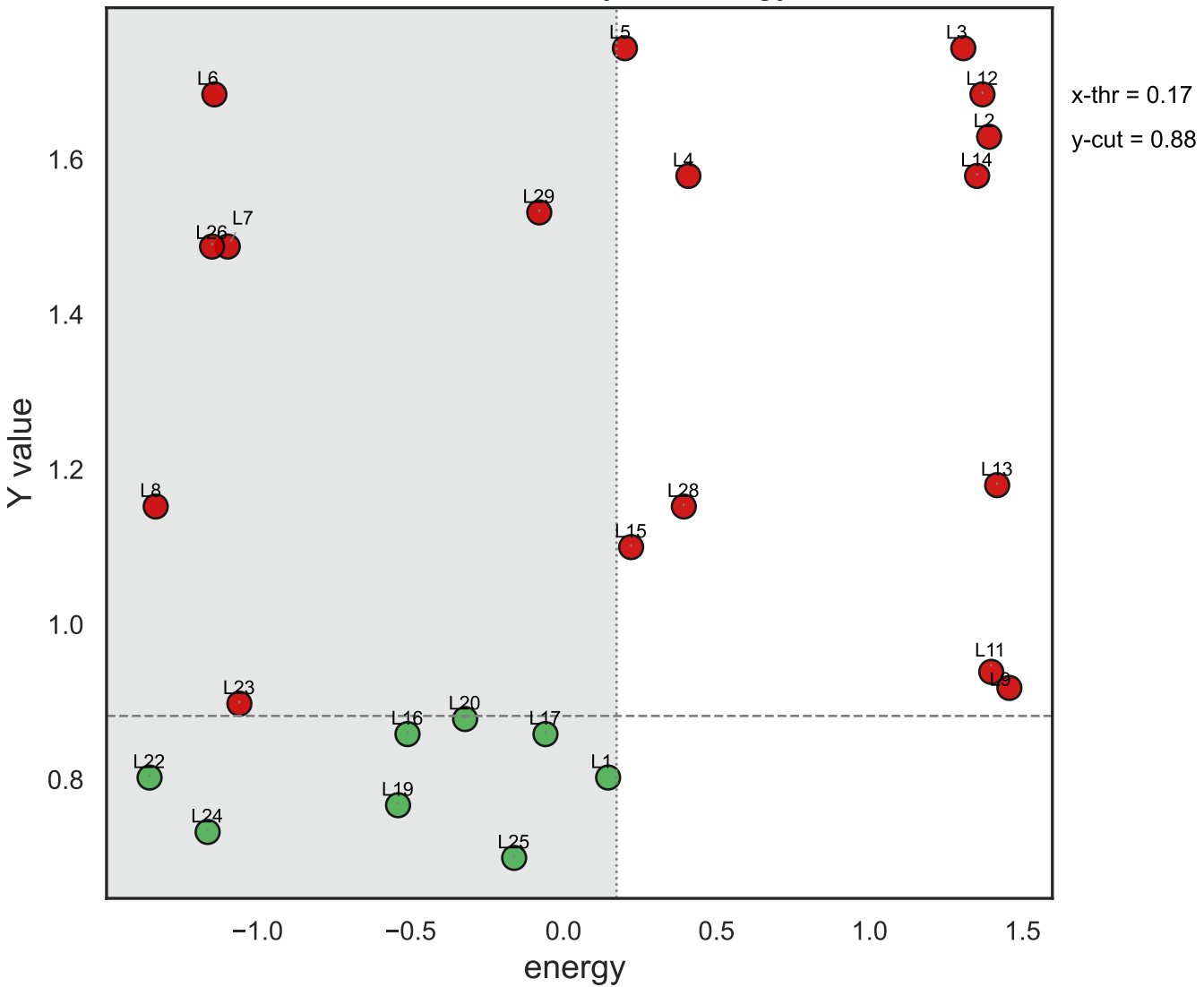
Threshold analysis: B1_B5_angle_5-2



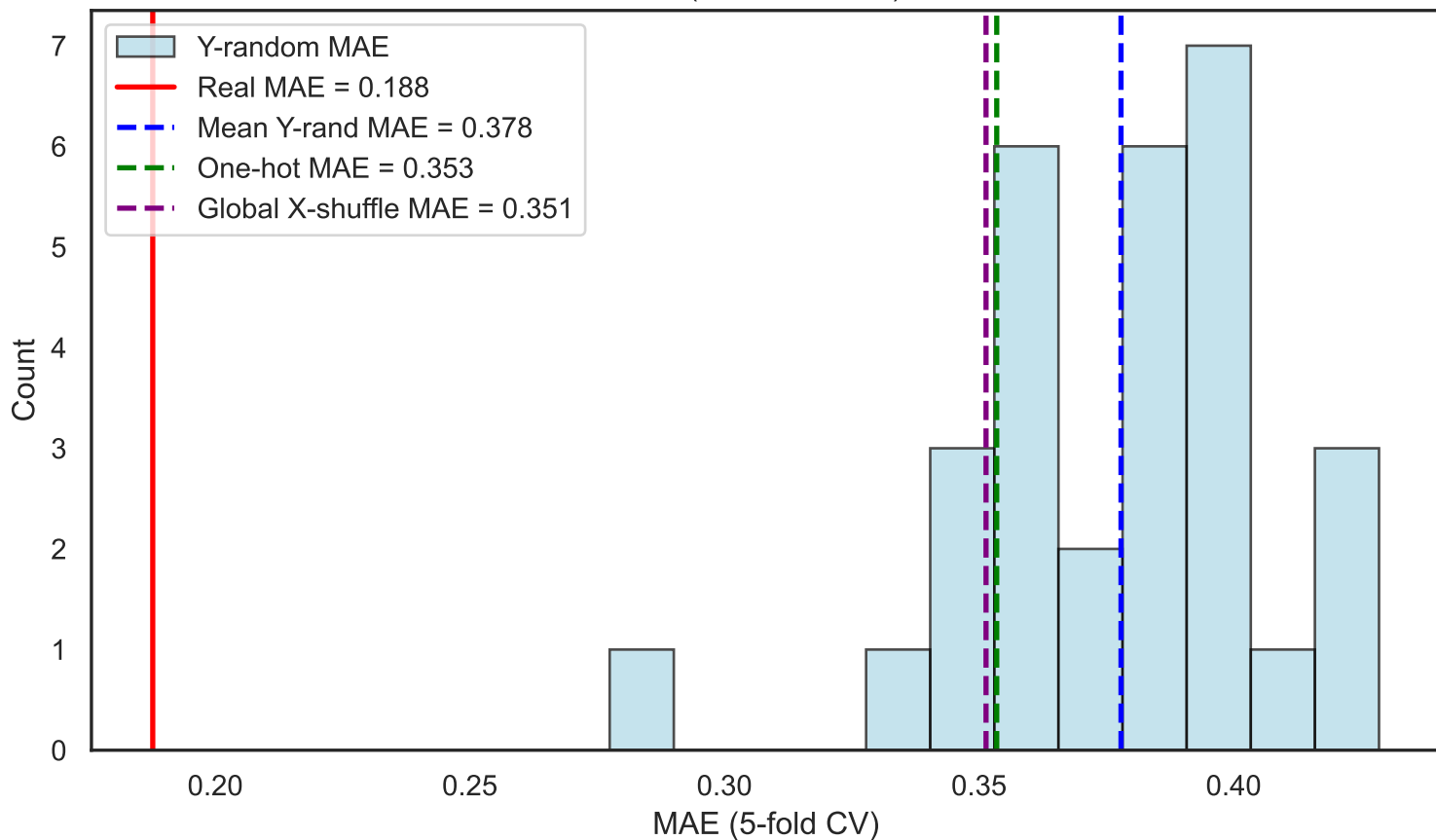
Threshold analysis: dihedral_[5, 1, 8, 6]



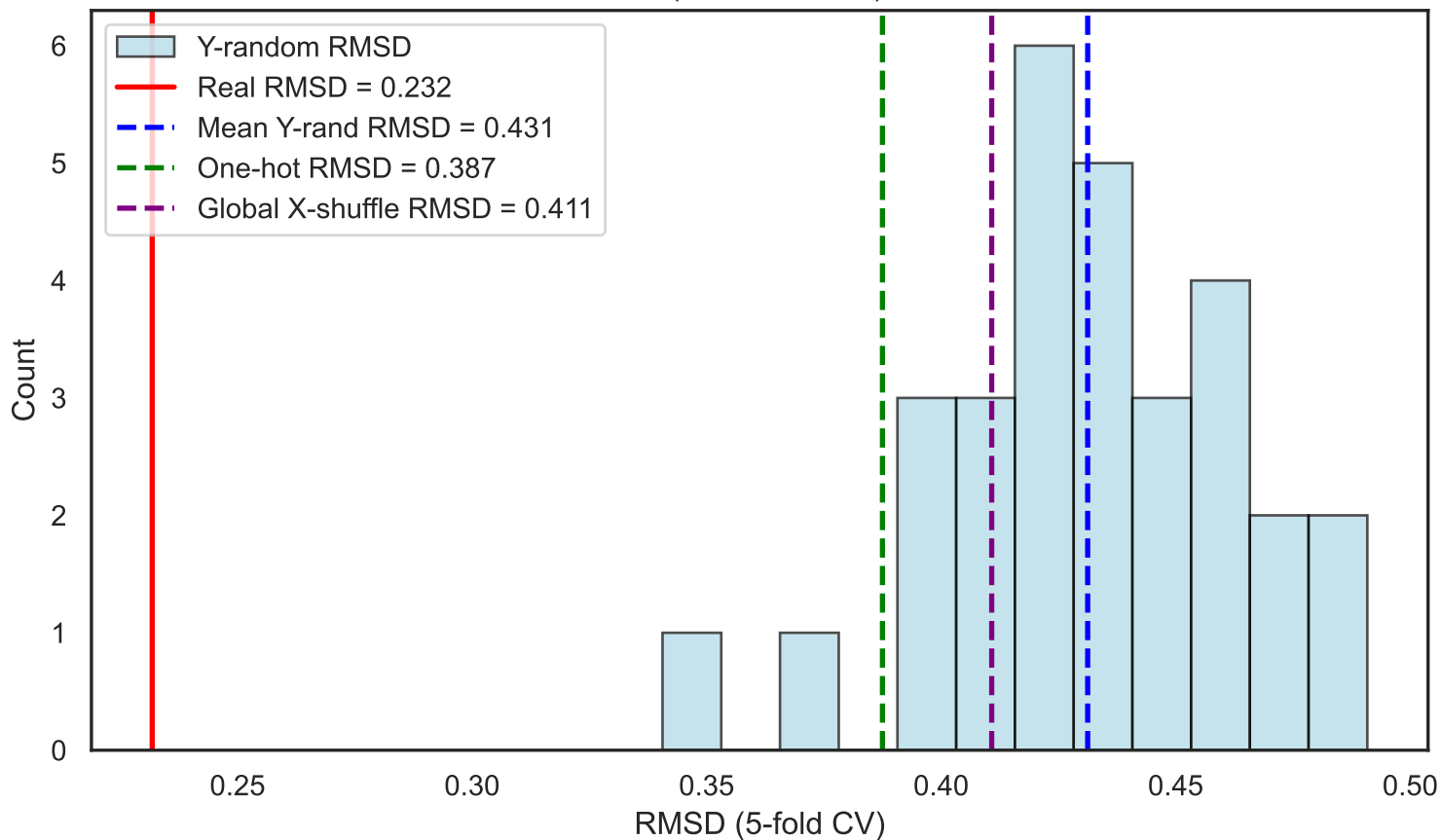
Threshold analysis: energy



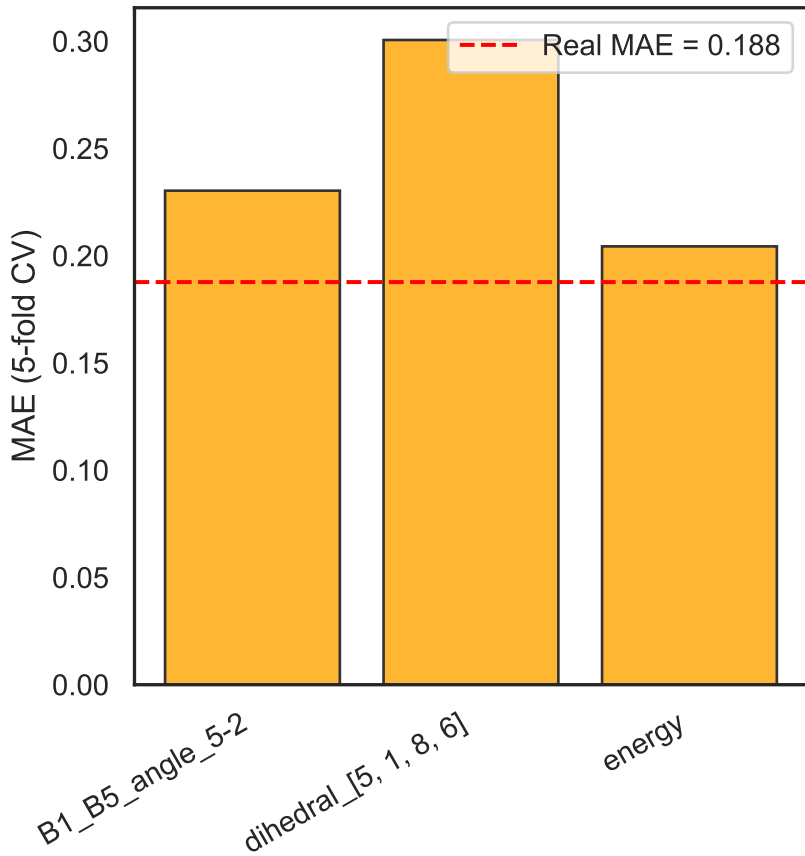
Validation (lower is better): MAE



Validation (lower is better): RMSD



Per-feature shuffle test: MAE (lower is better)



Per-feature shuffle test: RMSD (lower is better)

